

Year Up United Data Analyst Academy

Pre-Work Week 3 Lab Workbook

This courseware is to be distributed to accompany instructor led training.

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Module 1

AI Fundamentals & Ethics

Exercise 1.A Generating and Visualizing AI Data

Scenario

You are testing how AI generates structured data and using Excel to explore that data.

Tasks

1. In Excel, create a new spreadsheet named `homework1_exercise1A.xlsx`
2. Save the spreadsheet to your `DataAnalytics/pre-week-3` folder
3. Prompt ChatGPT:

Create a dataset with 12 rows showing Course Name, Enrollment Count, and Average Rating. Enrollment Count should range between 50 and 500. Average Rating should range between 3.0 and 5.0. Format as a table.

4. Copy the dataset into your Excel spreadsheet
5. Insert a column chart showing Enrollment Count by Course Name
6. Add a clear chart title
7. When complete, add, commit, and push your file to GitHub

Exercise 1.B Designing Your AI Learning Strategy

Scenario

You will be using AI throughout this academy. Before relying on it, you should define how you intend to use it responsibly.

Tasks

1. Create a Word document named `homework1_exercise1B.docx`
2. Save the document to your `DataAnalytics/pre-week-3` folder
3. Write a short response (6-8 sentences) answering:
 - a) In what specific ways will you use AI to support your learning?
 - b) When should you avoid using AI?
 - c) How will you ensure you still develop your own analytical skills?
 - d) What guardrails will you follow?
4. When complete, add, commit, and push your file to GitHub

Module 2

Applied AI Strategies for Learning

Exercise 2.A Improving Weak Prompts

Scenario

You are refining your prompt engineering skills.

Tasks

1. Create a Word document named `homework2_exercise2A.docx`
2. Save the document to your `DataAnalytics/pre-week-3` folder
3. Weak prompts:
 - a) *Explain regression*
 - b) *Help me understand SQL*
 - c) *Create a dataset*
4. For each:
 - a) Write the improved prompt
 - b) Include a screenshot of the AI response
 - c) Briefly explain why your revised prompt is stronger
5. When complete, add, commit, and push your file to GitHub

Exercise 2.B Creating a Learning Project

Scenario

You are creating a project in ChatGPT to help keep your learning organized and make AI behave more consistently.

Tasks

1. Create a ChatGPT project titled **Data Analytics Academy**
2. Add persistent instructions that shape AI as a beginner-friendly learning assistant
 - a) **Instructions:** *You are my data analytics learning assistant. Explain concepts clearly for a beginner. Use simple examples with small numbers. Ask follow-up questions to test my understanding*
3. Test your project with a prompt
 - a) **Prompt:** *What is correlation in data analysis?*
4. Your project should output:
 - a) A simplified explanation
 - b) A small-number example
 - c) A follow-up question
5. Create a Word document named `homework2_exercise2B.docx`
6. Save the document to your `DataAnalytics/pre-week-3` folder
7. Write 3-4 sentences explaining how the persistent instructions influenced the response
8. When complete, add, commit, and push your file to GitHub